PREP 1, LESSONS 1–2

Stores and the kinetic store

Energy stores and transfers; describing a transfer; kinetic energy

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons, 1 mark each

1 Define precision. DN 850 [1]

How close repeated measurements are to each other.

2 What is a control variable? DN 872 [1]

A variable kept the same to make the test fair.

3 What is an anomalous result? DN 895 [1]

A result that does not fit the pattern of the others.

Key ideas LESSONS 1–2

Show your working

4 A cyclist brakes and stops. Complete the sentence.

Energy is transferred mechanically (mechanically / electrically / by waves) from the kinetic (kinetic / chemical / thermal) store of the cyclist to the thermal (elastic / thermal / nuclear) store of the brakes. [2] All three ✓✓; two ✓

5 A ball of mass 0.50 kg moves at 4.0 m/s. Use $E_k = \frac{1}{2} m v^2$

Calculate the energy in the kinetic store of the ball. [3]

$E_k = \frac{1}{2} m v^2$ ✓

$E_k = \frac{1}{2} \times 0.50 \times 4.0^2$ ✓

$E_k = 4.0\,\text{J}$ ✓

6 Name the two quantities that kinetic energy depends on. [1]

Mass and speed ✓

7 A car's speed doubles. Its kinetic energy becomes [2 times] [4 times] [8 times] as big. [1]